

An Area Delay Optimized Carry-Select Adder

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Abstract

In arithmetic and logic unit of digital signal processor (DSP), adder is the important hardware unit. Carry select adder (CSLA) is the best example of the adder used in DSP and it is widely used in many data processors to increase speed. So, the adder performance affects the overall system-performance. The Regular Square root (SQRT) CSLA consists of two Ripple Carry Adders (RCA), so it consumes more area due to the presence of two Ripple Carry Adders (RCA) in the structure. To minimize the area of Regular SQRT CSLA, one of the RCAs was replaced by a Binary to Excess-1 Converter (BEC) with slight increase in delay. Many such techniques have provided to design variety of SQRT CSLAs by using Common Boolean Logic (CBL), First Addition Logic (FAL), Add-one circuit, Modified Reduced Logic Block (MRLB) etc. to achieve low area, delay and power. This work proposes a new design comprising of Carry generation (CG) and Carry Select (CS) to reduce the area as compared to the Regular SQRT CSLA and Modified SQRT CSLA using BEC. The proposed design is synthesized and simulated in Xilinx ISE design suite 14.2 and is implemented on Spartan 3E XC3S1600E-5-FG484 FPGA device. The comparison shows how the proposed SQRT CSLA is better than the existing regular SQRT CSLA and SQRT CSLA using BEC. The speed of proposed model is also enhanced for higher number of bits than the SQRT CSLA.

Keywords: CSLA, BEC, FPGA, SQRT, CBL

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INTRODUCTION

In VLSI technology area, delay and power are important factors when designing adder in digital circuits. Effective design of adder improves the performance of whole digital system. Adders are also used in memory fetched operation to compute physical address. There are different types of adders used in digital system. For example CSLA, RCA [1], Kogge Stone Adder (KSA) [2], Brent Kung Adder (BKA) [3], Carry Look-ahead Adder (CLA), Carry Save Adder (CSA), Carry Skip Adder (CSkA) [4] etc., but fast addition with low area and power is still challenging. In KSA and BKA, parallel prefix adder was designed by minimizing number of operation stages to reduced delay[5]. In carry skip adder, it uses skip logic to reduce a delay but number of gates required is more, so area has been increased. In carry look ahead adder, delay has been reduced by reducing the number of gates through which carry signals get propagated. In carry save tree, three bits are added parallel at a time and carry is stored in present stage and get added with next bits, so that delay has been

minimum in this scheme. In this paper, different approaches for carry select adder are used to reduced delay, area and power. Circuit can be designed by using two approaches. One is "System level" which reduce the path in order to optimize the critical path delay [7]. Another approach is "Circuit level", which is designed in transistor level to optimize the area, power or delay [8]. There are number of works implemented on design of high speed, low-power and area-efficient adders. Researchers use different algorithms[10] for addition and various structures for multi-bit adders. As we know in adder, speed of addition is inversely proportional to carry propagation time. RCA consist of Full Adders (FA) connected in cascade. So, output of next stage depends on the carry out bit of previous stage and delay increases. In regular SQRT CSLA[9], two ripple carry adders were used, so area was also increased. To avoid this drawback of regular CSLA, one RCA was replaced by BEC-1 convertor[6]. This reduces area with slight increase in delay. By analyzing above models, we designed our new

model to optimize area and delay by using Carry Generation (CG) and Carry Select (CS) unit instead of using two RCA units. CG unit is a combination of two units, one is CG_0 to generate carry output with input $C_{in}=0$ depends on input; and another is CG_1 for $c_{in}=1$. Output of CG unit is given to CS unit. According to input carry signal CS unit selects final carry output.

EXISTING SYSTEM

Regular/Conventional Adder

Theoretical area and delay are calculated by considering number of logic gates like AND, OR and NOT required to implement the circuit. The area and delay are calculated by adding the total number of AON gates required for each logic block and calculating the number of unit delay in longest path respectively[11]. Implementation for XOR

gate using AOI is as shown in Figure 1. For one XOR gate, two NOT, two AND and one OR gate are required.

Area Calculation of Regular SQRT CSLA

Architecture of regular 16-bit SQRT CSLA is shown in Figure 2. It has two sets of RCA. The 16-bit RCA with $C_{in}=0$ consists of four RCAs of different widths. Similarly, RCA with $c_{in}=1$ consists of four RCAs of different widths. Four 2:1 MUX select the final sum and carry. For 16 bit regular CSLA, 144 AOI gates are used as given below:

Gate calculation for 16 bit conv CSLA:

$$2 \text{ bit RCA} = (5+5) \times 2 = 20,$$

$$3 \text{ bit RCA} = (5+5+5) \times 2 = 30,$$

$$4 \text{ bit RCA} = (5+5+5+5) \times 2 = 40,$$

$$5 \text{ bit RCA} = (5+5+5+5+5) \times 2 = 50,$$

$$\text{MUX} = 4, \text{ and}$$

$$\text{Total} = 144.$$

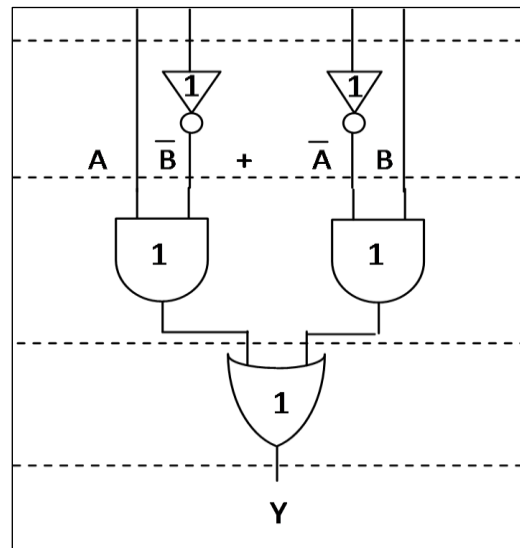


Fig. 1: Implementation of XOR Gate.

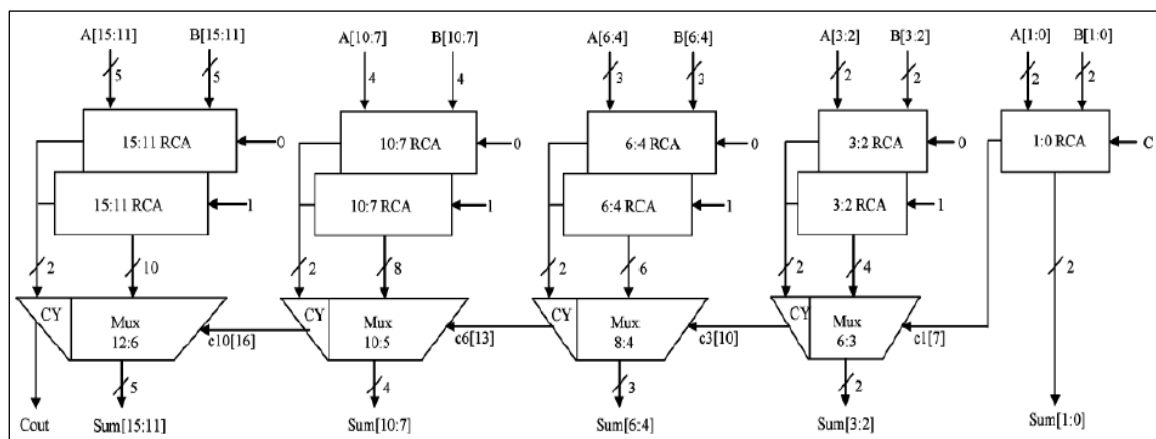


Fig. 2: 16 bit Regular CSLA.

PROPOSED ADDER DESIGN

Proposed CSLA

Structure of proposed CSLA is shown in Figure 3. It is consist of four main blocks: (1) Half Sum Generation (HSG), (2) Carry Generation (CG)- two sub units: (a) CG0 (generate a carry 0 signal) (b) CG1 (generate carry 1 signal), (3) Carry Select (CS) unit, and last one is (4) Full Sum Generation (FSG). Each unit is formed by using a basic logic gates.

Area Calculation of Proposed CSLA

For 16 bit proposed CSLA, area calculation is given below. A HSG unit contains 28 logic gates, CG unit is formed by using a total 44 logic gates, CS unit contains 28 logic gates and FSG unit takes 14 logic gates.

Gate calculation for 16 bit prop CSLA:

HSG unit=28,

CG0 unit=24,

CG1 unit=20,

CS unit=28,

FSG unit=14,

2 bit RCA=20,

Total=134.

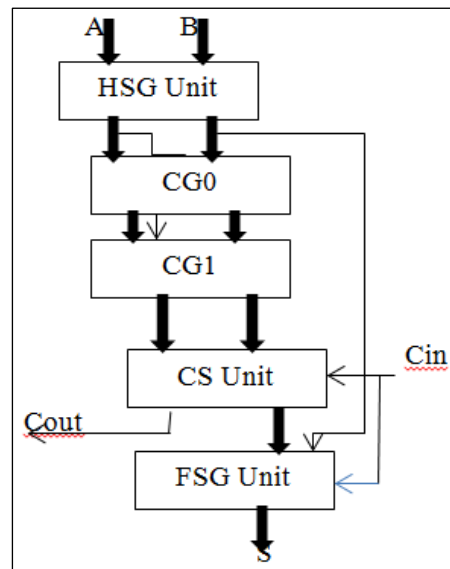


Fig. 3: Structure of Proposed CSLA.

SYNTHESIS RESULTS

In synthesis result, report of number of LUTs used, memory used and number of slice registers used are given. Figures 4 and 5 show the summary of resources used by regular and proposed CSLA respectively.

Device Utilization Summary				
Slice Logic Utilization	Used	Available	Utilization	Note(s)
Number of Slice Registers	0	18,224	0%	
Number of Slice LUTs	18	9,112	1%	
Number used as logic	18	9,112	1%	
Number using O6 output only	10			
Number using O5 output only	0			
Number using O5 and O6	8			
Number used as ROM	0			
Number used as Memory	0	2,176	0%	
Number of occupied Slices	11	2,278	1%	
Number of MUXC's used	0	4,556	0%	

Fig. 4: Regular CSLA 16 bit Report.

Device Utilization Summary				
Slice Logic Utilization	Used	Available	Utilization	Note(s)
Number of Slice Registers	0	126,800	0%	
Number of Slice LUTs	16	63,400	1%	
Number used as logic	16	63,400	1%	
Number using O6 output only	7			
Number using O5 output only	0			
Number using O5 and O6	9			
Number used as ROM	0			
Number used as Memory	0	19,000	0%	
Number used exclusively as route-thrus	0			
Number of occupied Slices	8	15,850	1%	

Fig. 5: PROP CSA 16 bit Report.

PERFORMANCE COMPARISON

As shown in Table 1, number of LUTs required by prop CSLA is very less as compared to conventional CSLA. But one limitation is that as adder width increases, the delay also increases.

Table 1: Result.

Width	Adder	Area (LUTs)	Delay (ns)
16	CONV CSLA	18	5.9
	PROP CSLA	16	5.07
32	CONV CSLA	39	7.99
	PROP CSLA	32	9.57
64	CONV CSLA	85	9.30
	PROP CSLA	64	18.6

COMPARISON AMONG DIFFERENT BIT WIDTH CSLAs

Following Figures 6 and 7 show comparison of proposed and regular CSLA for area, delay and power. As shown, area is calculated in terms of number of LUTs and delay is calculated in nanoseconds.

CONCLUSION

We have identified the redundant logic operations involved in conventional and BEC based CSLA to optimize area, delay and power. We have eliminated all the redundant logic operations which are involved in conventional and BEC based CSLAs and designed a proposed CSLA with minimum logic operations. In the proposed scheme,

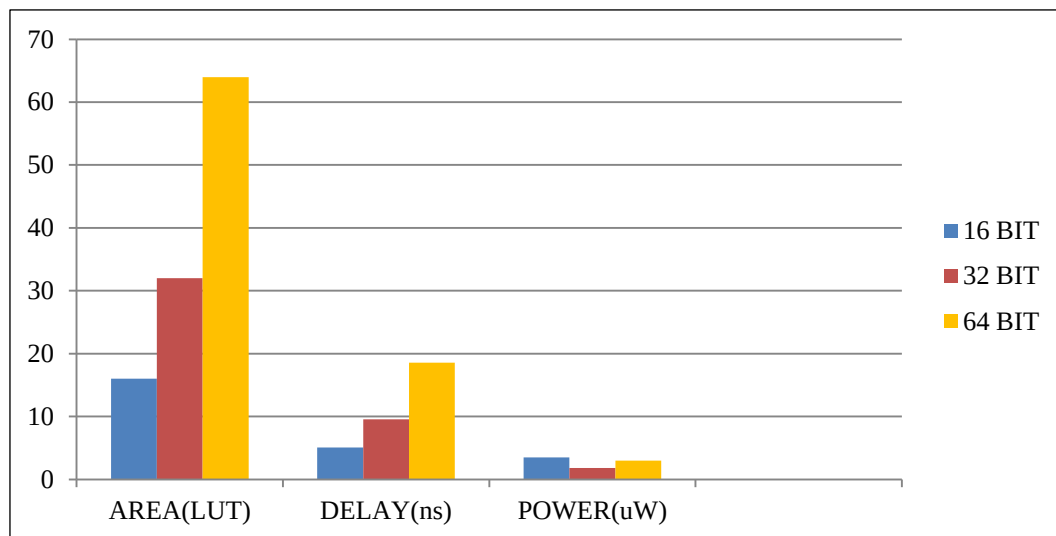


Fig. 6: Comparison of Proposed CSLA.

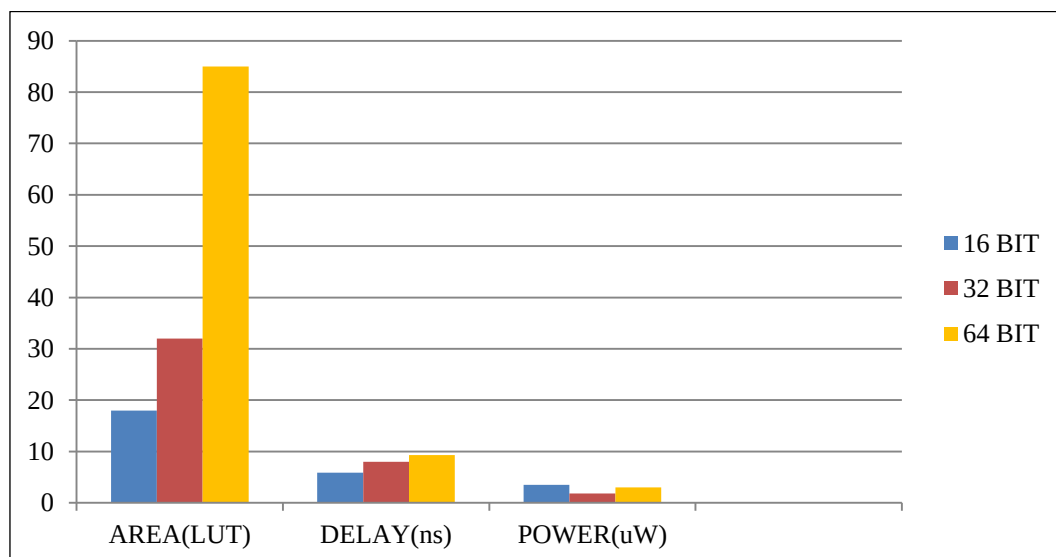


Fig. 7: Comparison of Regular CSLA.

carry is generated before final sum calculation by using CS unit, so that final sum does not depend on carry bit of previous stage and delay is reduced in proposed CSLA. Carry bits generated by CS unit follow specific bit pattern, which is used for logic optimization of CS unit. So, proposed CSLA involves less area and delay as compared to conventional CSLA.

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